Hogyun **Kim**

Ph.D. Student · Multi-Robot Systems

100, Inha-ro, Michuhol-gu, Incheon, Republic of Korea

(+82) 10-9364-2307 | hg.kim@inha.edu | hogyun2.github.io | hogyun2 | hogyun-kim | Google Scholar

"Scalable and Collaborative Autonomy for Multi-Robot Systems"

Summary

A Ph.D. student in the M.S.–Ph.D. integrated program at SPARO Lab, Inha University, advised by Prof. Younggun Cho. Researching scalable and collaborative autonomy for multi-robot systems: how robots perceive, map, and operate in shared environments through communication-efficient coordination. Current focus is on communication-aware multi-robot SLAM and map merging, and on 3D foundation models for collaborative embodied agents.

Highlights

- First-author papers in IJRR, IEEE T-FR, IEEE RA-L, IEEE/RSJ IROS, and IEEE ICRA on multi-robot LiDAR SLAM, map merging, and LiDAR/radar/sonar place recognition.
- Awards: Outstanding WiRA Student Paper Award (IEEE ICRA 2026), Best Paper on Robot Perception Finalist (IEEE ICRA 2026), Qualcomm Innovation Fellowship 2026 Finalist, and Samsung HumanTech Paper Award (Silver Prize).
- Challenges: 2nd place in the SLAM category of the Hilti-Trimble SLAM Challenge (ICRA 2026) and 2nd among 131 teams in the TartanIMU Challenge (IROS 2026).
- Principal investigator of an NRF Ph.D. Candidate Research Fellowship on real-time multi-robot SLAM in communication-constrained scenarios.

Education

Inha University — *Incheon, Republic of Korea*

M.S.–Ph.D. Integrated in Electrical and Computer Engineering — *Mar. 2023 - Current*

- Advisor: Prof. Younggun Cho

Inha University — *Incheon, Republic of Korea*

B.S. in Naval Architecture and Ocean Engineering — *Mar. 2017 - Feb. 2023*

- Minor: Information and Communication Engineering
- Inha Naval Architecture and Ocean Engineering Alumni Association Scholarship, given to promising students in the department

Work Experience

Undergraduate Researcher (Internship) — *Incheon, Republic of Korea*

Spatial AI and Robotics (SPARO) Lab, Inha University — *Oct. 2021 - Feb. 2023*

- Advisor: Prof. Younggun Cho

Research Projects

Building a Real-Time Multi-Robot SLAM Framework in Communication-Constrained Scenarios — *National Research Foundation of Korea*

Principal Investigator, NRF Ph.D. Candidate Research Fellowship — *Sep. 2025 - Current*

- Multi-robot SLAM framework that keeps robots collaborating under delays, dropouts, and limited bandwidth while exchanging only the minimum necessary information
- Year 1: lossless information-exchange policy on an exchange graph, built on lightweight global descriptors and sequence matching
- Year 2: submap-based inter-robot registration and distributed back-end optimization for globally consistent mapping
- Year 3: datasets and hardware with real communication logs, full-system integration, and field validation

Stereo Vision-based Object Localization for Unmanned Surface Vehicles — *Samsung Heavy Industries*

Participant, Industry Project — *2023*

- Developed *Obcalizer*, a training-free stereo-vision framework for maritime metric object localization (detection, metric depth and 3D position estimation, and data association)
- Open-vocabulary detection with YOLO-World, box-level epipolar context matching for metric depth, and a depth-aware Kalman filter for tracking
- Lowest average 3D position error on real-world USV datasets (1.52 m vs. 3.51 m for the strongest baseline); runs at the 10 Hz sensor rate on an NVIDIA Jetson AGX Orin

TartanIMU Challenge @ IROS 2026 *IROS 2026 Workshop*

Team Member, Challenge *Oct. 2026*

- *Single-Head IMU Velocity Estimation Across Platforms via a Multi-Scale, Embodiment-Aware Encoder*: one network estimating velocity from IMU alone across cars, quadrupeds, drones, and handheld devices
- **2nd place among 131 teams**

Hilti-Trimble SLAM Challenge 2026 @ ICRA 2026 *ICRA 2026 Workshop*

Team Member, Challenge *Jun. 2026*

- $\sqrt{\text{VINS}}$*-Based 360 Visual-Inertial SLAM Systems*: a 360-camera visual-inertial SLAM pipeline for long indoor sequences with dark regions, repetitive corridors, and dynamic objects
- **2nd place in the SLAM category**

Awesome LiDAR Place Recognition [★ 224] *Personal Project*

Maintainer, Open Source *2024 - Current*

- A curated list of LiDAR place recognition research with code or download pages, covering handcrafted and learning-based methods, single-/multi-session and multi-robot datasets, and related multi-robot mapping and SLAM algorithms

Publications

Underline denotes co-first authors or co-corresponding authors.

International Journal

1. **Hogyun Kim**, Juwon Kim, Dongjin Cho, Jiwon Choi, Geonmo Yang, Seokhwan Jeong, Jiyun Lee, Miryeong Park, Hyungtae Lim, and Younggun Cho, "SKiD-SLAM: Resource-Aware Distributed Multi-Robot LiDAR SLAM for Planetary Missions with Field Deployments at Analog Sites," *IEEE Trans. Field Robot. (T-FR)*, 2026. Accepted. To appear.
2. **Hogyun Kim**, Jiwon Choi, Juwon Kim, Geonmo Yang, Seokhwan Jeong, Hyungtae Lim, and Younggun Cho, "Commerge: Communication-Efficient, Robust and Fast LiDAR Map Merging Framework for Multi-Robot Coordination in Resource-Constrained Scenarios," *Int. J. Robot. Res. (IJRR)*, 2026. Accepted. To appear. **Samsung HumanTech Paper Award (Silver Prize)**.
3. Gilhwan Kang, **Hogyun Kim**, Byunghee Choi, Seokhwan Jeong, Young-Sik Shin, and Younggun Cho, "Uni-Mapper: Unified Mapping Framework for Multi-modal LiDARs in Complex and Dynamic Environments," *IEEE Trans. Intell. Veh. (T-IV)*, Jul. 2025.
4. **Hogyun Kim**, Byunghee Choi, Euncheol Choi, and Younggun Cho, "ReFeree: Radar-Based Lightweight and Robust Localization Using Feature and Free Space," *IEEE Robot. Automat. Lett. (RA-L) with ICRA*, Oct. 2024.
5. **Hogyun Kim**, Jiwon Choi, Taehu Sim, Giseop Kim, and Younggun Cho, "Narrowing your FOV with SOLiD: Spatially Organized and Lightweight Global Descriptor for FOV-constrained LiDAR Place Recognition," *IEEE Robot. Automat. Lett. (RA-L) with ICRA*, Aug. 2024.
6. Seokhwan Jeong, **Hogyun Kim**, and Younggun Cho, "DiTer: Diverse Terrain and Multi-Modal Dataset for Field Robot Navigation in Outdoor Environments," *IEEE Sensors Letters*, Jan. 2024.

International Conference

1. **Hogyun Kim**, Jiwon Choi, Jungwoo Lee, and Younggun Cho, "RadLoc: Radar-based 3-DoF Global Localization via Fast, Robust, and Lightweight Spatial Descriptor Across Diverse Environmental Scenarios," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Pittsburgh, USA, 2026. **IROS 2026 Travel Support**.
2. Jiwon Choi, **Hogyun Kim**, Geonmo Yang, Juhui Lee, and Younggun Cho, "KISS-IMU: Self-supervised Inertial Odometry with Motion-balanced Learning and Uncertainty-aware Inference," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Vienna, Austria, 2026. **Outstanding WiRA Student Paper Award; Best Paper Award on Robot Perception Finalist; Qualcomm Innovation Fellowship 2026 Finalist**.
3. Seokhwan Jeong, **Hogyun Kim**, and Younggun Cho, "MARSCalib: Multi-robot, Automatic, Robust, Spherical Target-based Extrinsic Calibration in Field and Extraterrestrial Environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Hangzhou, China, 2025.
4. Juwon Kim, **Hogyun Kim**, Seokhwan Jeong, Youngsik Shin, and Younggun Cho, "DiTer++: Diverse Terrain and Multi-modal Dataset for Multi-Robot SLAM in Multi-session Environments," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Atlanta, USA, 2025.

5. Jiwon Choi, Dongjin Cho, Gihyeon Lee, **Hogyun Kim**, Geonmo Yang, Joowan Kim, and Younggun Cho, "Po-LaRIS Dataset: A Maritime Object Detection and Tracking Dataset in Pohang Canal," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Atlanta, USA, 2025.
6. **Hogyun Kim**, Gilhwan Kang, Seokhwan Jeong, Seungjun Ma, and Younggun Cho, "Robust Imaging Sonar-based Place Recognition and Localization in Underwater Environments," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, London, UK, May 2023.

Workshop & Preprint

1. **Hogyun Kim**, Jiwon Choi, Juwon Kim, Geonmo Yang, Dongjin Cho, Hyungtae Lim, and Younggun Cho, "SKiD-SLAM: Robust, Lightweight, and Distributed Multi-Robot LiDAR SLAM in Resource-Constrained Field Environments," in *ICRA Workshop on Field Robotics*, Atlanta, USA, May 2025. **Spotlight Talk**.
2. Byunghee Choi, **Hogyun Kim**, and Younggun Cho, "ReFeree: Radar-based Efficient Global Descriptor using a Feature and Free Space for Place Recognition," in *ICRA Workshop on Radar in Robotics*, Yokohama, Japan, May 2024.

Domestic Journal

1. Seokhwan Jeong, **Hogyun Kim**, Juwon Kim, and Younggun Cho, "Spherical Target-based LiDAR-Camera Extrinsic Calibration for Multi-Robot System in Field Environments," *J. Korean Inst. Inf. Technol.*, Aug. 2025.
2. Jiwon Choi, Taehu Sim, Gihyeon Lee, Jaeyong Lee, Gilhwan Kang, **Hogyun Kim**, and Younggun Cho, "Study on Efficient Multi-floor Navigation Using a Visual Marker and Floor Plan Map," *J. Inst. Control, Robot. Syst. (ICROS)*, Jan. 2024.

Honors & Awards

International

2026	**Outstanding WiRA Student Paper Award (KISS-IMU, 2nd author),** IEEE ICRA 2026	*Vienna, Austria*
2026	**Best Paper Award on Robot Perception Finalist (KISS-IMU, 2nd author),** IEEE ICRA 2026	*Vienna, Austria*
2026	**2nd Place, SLAM Category,** Hilti-Trimble SLAM Challenge, ICRA 2026	*Vienna, Austria*
2026	**2nd Place among 131 Teams,** TartanIMU Challenge, Beyond Exteroception Workshop, IROS 2026	*Pittsburgh, USA*
2026	**Travel Support,** IEEE/RSJ IROS 2026	*Pittsburgh, USA*
2025	**Spotlight Talk,** ICRA 2025 3rd Workshop on Field Robotics	*Atlanta, USA*
2024	**Best Research Award (3rd Prize),** ICRA 2024 Workshop on Future of Construction	*Yokohama, Japan*

Domestic

2026	**Finalist (KISS-IMU),** Qualcomm Innovation Fellowship 2026 South Korea	*S.Korea*
2026	**Silver Prize,** 32nd Samsung HumanTech Paper Award, Samsung Electronics	*Seoul, S.Korea*
2026	**Graduate Research Award,** Inha University	*Incheon, S.Korea*
2026	**Research Presentation Award,** Inha University	*Incheon, S.Korea*
2025	**Ph.D. Candidate Research Fellowship,** National Research Foundation of Korea (NRF)	*Daejeon, S.Korea*
2025	**Excellent Research Award,** Korean Institute of ITS Conference	*Jeju, S.Korea*
2025	**Outstanding Research Award (Most-Applied Patent),** Inha University	*Incheon, S.Korea*
2025	**Outstanding Research Award (Excellence of Journal Publication),** Inha University	*Incheon, S.Korea*
2021	**Participation Award,** Autonomous Car Competition for College Students	*Hwaseong, S.Korea*
2021	**Control Prize in Design/Making,** Korea Autonomous Boat Competition (KABOAT)	*Changwon, S.Korea*
2021	**6th Prize,** Korea Autonomous Boat Competition (KABOAT)	*Changwon, S.Korea*
2021	**Vertically Integrated Project Award (1st Prize),** Inha University	*Incheon, S.Korea*
2020	**Vertically Integrated Project Award (1st Prize),** Inha University	*Incheon, S.Korea*
2020	**1st Prize in Autopilot,** Korea Autonomous Boat Competition (KABOAT)	*Changwon, S.Korea*
2020	**4th Prize,** Korea Autonomous Boat Competition (KABOAT)	*Changwon, S.Korea*

Presentations

2025	**Invited Talk,** KRoC Organized Session on Cutting-Edge Tech. and App. for Autonomous Multi-Robot/Agent Collaboration	*S.Korea*
Jul. 2024	**"Ranging-sensor-based Lightweight Place Description for Distributed SLAM",** IEEE RAS Summer School on Multi-Robot Systems	*Prague, Czech Republic*
Jul. 2024	**"Radar-based Place Recognition Using a Robust and Lightweight Global Descriptor for Limited Environment Multi-Robot Cooperation",** ICROS Organized Session on Robotics and AI Tech. for Space and Extreme Environments	*Daejeon, S.Korea*
May 2024	**"ReFeree: Radar-based Efficient Global Descriptor using a Feature and Free Space for Place Recognition",** ICRA 2024 Workshop on Radar in Robotics	*Yokohama, Japan*
Oct. 2022	**"Robust Sonar-based Place Recognition in Underwater Environments",** IEEE/RSJ IROS 2022 Late-Breaking Results	*Kyoto, Japan*

Patents

International

2025	**Lightweight Robot Localization for LiDARs with Limited FOV**	*Incheon, S.Korea*
2025	**Radar-based Lightweight Robot Localization**	*Incheon, S.Korea*

Domestic

2024	**Lightweight Robot Localization for LiDARs with Limited FOV**	*Incheon, S.Korea*
2024	**Radar-based Lightweight Robot Localization**	*Incheon, S.Korea*
2024	**Sensor Extrinsic Calibration in Limited Environments**	*Incheon, S.Korea*
2023	**A Method on the Efficient Multi-Floor Navigation Using Visual Marker and Floor Plan Map**	*Incheon, S.Korea*

Academic Services

2026	**Student Representative,** Organizing Committee, IEEE SSRR 2026	*Incheon, S.Korea*
2026	**Reviewer,** IEEE RA-L, IEEE ICRA, IEEE T-ASE, IEEE IoT-J, Autonomous Robots	
2025	**Reviewer,** IJRR, IEEE RA-L, IEEE/RSJ IROS, IEEE JOE	
2024	**Reviewer,** IEEE RA-L	

Training

2025	**Hands-on Training on the SPARUS II AUV Platform,** IQUA Robotics	*Girona, Spain*
2024	**IEEE RAS Summer School on Multi-Robot Systems,** Czech Technical University	*Prague, Czech Republic*

Skills

Programming C++, Python, MATLAB, LaTeX
Tools ROS, Docker